VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

BILLET-F5 Eight-Microphone Acoustic Speech Frontend

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 05:45 UTC+8 Beijing Time)

Abstract

BILLET-F5 combines programmable eight-microphone fractional-delay beamforming, normalized LMS acoustic echo cancellation, spectral voice-activity detection and voice-gated 64-band Log-Mel tokens. A separate acoustic pipeline preserves the inherited four-channel FIR, SOS IIR, polyphase resampling and spectral interfaces. Software-vs-Hardware numerical comparison used an independent integer Python oracle against simulated RTL, not physical FPGA acoustic execution. Across three seeds and four AEC tap counts, evaluation echo reduction was at least 58.43 dB and recovery reduction at least 58.37 dB. All twelve final double-talk windows reported 100% freezing with exact public audio/freeze agreement. Broadside SNR improved 8.92–9.23 dB. A continuously ready, 32-tap, four-frame simulation window accepted 2,048 tuples over 4,876,126 clocks, approximately 42,001 tuples/s at 100 MHz. Evidence: E069–E071. L1 RTL timing lint passed across 24 RTL files. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA closed the 10.0 ns baseline with WNS_setup=+1.808 ns and WNS_hold=+0.040 ns. L3 physical Arria 10 FPGA HIL passed three consecutive deterministic bridge-connectivity probes; these did not execute acoustic workload vectors. Evidence: E077–E079.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

billet_f5_top retains the literal billet_f4_top u_legacy instance and adds an independent mono acoustic path. Parallel microphone interpolation and registered reductions bound beamformer logic depth. Time-multiplexed AEC arithmetic trades sample latency for resource reuse, using a sequential restoring divider rather than combinational division. An acknowledged audio fork delivers each sample exactly once to audio and spectrum consumers. Spectrum powers feed both VAD and Mel; retained frame ownership prevents stalled decisions or tokens from crossing frame boundaries. Blocks are arranged row-major for a three-column diagram. Sources: E001, E023–E027.

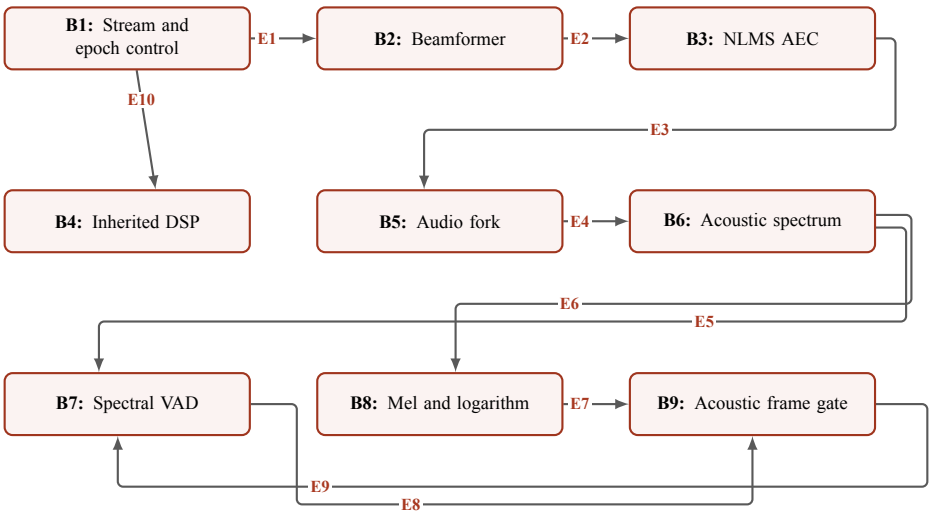


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Stream and epoch control	Accepts synchronized tuples, stages eight delays and commits acoustic modes after old-epoch work drains. Legacy control remains independent.
B2: Beamformer	u_beamformer holds microphone histories and delay banks. Eight stallable stages interpolate, average and align far-end metadata.
B3: NLMS AEC	u_aec holds weights, far history and smoothed powers. u_divider performs 56 restoring iterations per tap update.
B4: Inherited DSP	u_legacy retains four-channel FIR/SOS processing, coefficient banks, SRC, spectrum, Mel and logarithm engines.
B5: Audio fork	u_audio_fork stores post-AEC audio and freeze metadata with independent acknowledgements for its two destinations.
B6: Acoustic spectrum	u_acoustic_spectrum collects 512 samples and generates 257 windowed real-spectrum powers using registered direct-transform arithmetic.
B7: Spectral VAD	u_vad accumulates bins 4–128, applies hysteresis and hangover, and retains the frame decision independently of event acknowledgement.
B8: Mel and logarithm	u_acoustic_mel processes every spectrum frame into 64 energies; u_acoustic_log produces inherited 16-bit token encoding.
B9: Acoustic frame gate	Forwards or discards whole token frames. Event completion and token retirement jointly release frame ownership.

Directed relationship	Meaning
E1: B1 to B2	Tuples, delays and epoch control
E2: B2 to B3	Beam sample and aligned far reference
E3: B3 to B5	Audio and freeze metadata
E4: B5 to B6	Acknowledged mono samples
E5: B6 to B7	Spectrum powers and bin indices
E6: B6 to B8	All 257 spectrum powers
E7: B8 to B9	64 ordered logarithmic tokens
E8: B7 to B9	Held decision and event completion
E9: B9 to B7	Frame-retirement acknowledgement
E10: B1 to B4	Independent inherited interface

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single clock and active-low asynchronous reset. Transfers occur on rising edges with pre-edge valid and ready high. New public outputs are zero during reset.
a_mics, a_far, a_valid, a_ready	Input; a_ready output	128; signed 16; 1; 1 bits	One synchronized tuple: microphone i occupies a_mics[16*i+:16] as signed PCM; a_far belongs to the same accepted sample index.
a_cfg_valid, a_cfg_ready, a_cfg_mic, a_cfg_delay	Input; a_cfg_ready output	1; 1; 3; DELAY_W+8 bits	Writes inactive delay entries. Low eight delay bits encode fraction f; upper bits encode integer k. Invalid k holds ready low.

Signal / group	Direction	Width	Contract
a_commit_valid, a_commit_ready, a_aec_enable, a_gate_enable	Input; a_commit_ready output	1 bit each	Commit requires eight written entries and acoustic drain. Enable values are captured on acceptance; an asserted request blocks new acoustic tuples and writes.
audio_valid, audio_ready, audio_data, audio_frozen	Output; audio_ready input	1; 1; signed 16; 1 bits	Post-AEC sample and transaction freeze decision. Disabled AEC returns beam audio with frozen=1.
a_t_valid, a_t_ready, a_t_data, a_t_band, a_t_last	Output; a_t_ready input	1; 1; 16; 6; 1 bits	Ordered acoustic bands 0–63; last marks band63 only. Gating suppresses complete inactive frames.
vad_valid, vad_ready, vad_active, vad_energy	Output; vad_ready input	1; 1; 1; 48 bits	One event per complete frame, including suppressed frames. Payload remains stable while stalled.
s_data, s_channel, s_valid, s_ready; m_data, m_channel, m_valid, m_ready	Inherited input/output streams	Signed 16-bit data; 2-bit channel; 1-bit handshakes	Unchanged four-channel legacy sample interfaces connected through u_legacy.
cfg_*, commit_*; sos_cfg_*, sos_commit_*	Inherited configuration inputs and ready outputs	FIR data signed16, address ADDR_W; SOS data signed18, section2, address3; channel2; handshakes1	Banked FIR/SOS coefficient writes and per-channel commits retain inherited completeness and priority rules.
chain_valid, chain_ready, chain_mode, chain_sections; feature_valid, feature_ready, feature_mode	Input; ready signals output	Handshakes1; chain fields3; feature_mode2 bits	Selects inherited chain ordering, active section count and feature mode independently of acoustic operation.
src_cfg_*, src_commit_*, src_l, src_m	Inherited configuration inputs and ready outputs	Data signed18; channel2; phase3; tap SRC_AW; L/M4; handshakes1	Configures polyphase coefficients and interpolation/decimation ratios, each 1–8.
p_valid, p_ready, p_power, p_channel, p_bin, p_last; t_valid, t_ready, t_data, t_channel, t_band, t_last; feature_error	Power input; token/status output; stream readiness opposite	Power32; bin9; token16; channel2; band6; other signals1	Inherited external-power processing, token output and malformed-descriptor error reporting. Exact declarations: rtl/billet_f5_top.sv, E025.

Parameter	Default	Meaning / supported values
TAP_COUNT	32	Inherited FIR taps: 1, 8, 16, 32 or 64.
MAC_LANES	1	Inherited FIR lanes: 1, 2 or 4.
SOS_COUNT	2	Inherited cascaded sections: integers 1–4.
SRC_TAPS	32	Inherited resampler taps: 1, 8, 16, 32 or 64.
DELAY_DEPTH	8	1, 8, 17 or 32 previous tuples per microphone; 0<= k<DELAY_DEPTH.

Parameter	Default	Meaning / supported values
AEC_TAPS	16	Adaptive taps: 1, 8, 16 or 32.
Derived widths	ADDR_W=5, SRC_ AW=5, DELAY_W=3	max(1,clog2(associated depth)). Unsupported parameters use a diagnostic branch and terminate simulation with PARAMETER ERROR.

III Behavior and corner cases

A Fractional-delay arithmetic

RNE(z,s) divides signed integer z by 2^s, rounding nearest with ties to even; satW denotes signed W-bit saturation. For each microphone, u=x[n−k], v=x[n−k−1], and b_i=sat16(RNE((256−f)u+fv,8)). The beam sample is sat16(RNE(sum b_i,3)); individual rounding precedes averaging. Missing history is zero, and history advances only on tuple transfers. Explicit ring wrapping supports depth17.

B Normalized cancellation

Signed24 weights represent w/2^22. Current-inclusive far history forms D=65536+sum x_j^2 in 36 bits and a signed45 dot product. Prediction y=sat16(RNE(sum w_j*x_j,22)); error e=sat16(b−y). Desired/far powers use P_new=P−(P>>5)+(sample²>>5). Freeze occurs when Ex_new<65536 or Ed_new>2*Ex_new+65536. Otherwise w_new=sat24(w−RNE(w,16)+trunc_toward_zero(e*x_j*2^21/D)). Updates use the same error and old weights, with no premature delta clipping. RTL inhibits weight writes when frozen, including leakage. Disabled AEC advances history/powers but returns b.

C Spectrum and token encoding

Non-overlapping 512-sample frames use symmetric Hann coefficients RNE(65536*(1−cos(2*pi*n/511))/2,0) . Window products use RNE shift16 and signed16 saturation. Signed Q16 cosine/negative-sine transform sums use RNE shift25 and signed18 saturation; squared magnitude saturates to unsigned32. Sixty-four HTK-Mel triangles use 66 breakpoints f_j=700*((1+8000/700)^(j/65)−1), evaluated at 16000*k/512 and quantized to Q16. Energy uses RNE shift16 into unsigned48. The inherited logarithm uses 17 log2 endpoints, four interval bits and twelve residual bits. Subtracting 30*65536, multiplying by 197283 with RNE shift16, and clipping to [−96*65536,0] precede token normalization by 21845 with RNE shift21. Tokens saturate to65535; zero energy yields zero.

D VAD and ownership

Energy sums inclusive bins4–128. E>=65536 activates the frame and sets hangover3. Previously active frames with E>=32768 retain activity and hangover; lower energies consume remaining hangover before inactivity. Every frame emits an event. Gate-disabled operation forwards all tokens. Event/token retirement preserves frame association; VAD does not independently freeze AEC.

E Timing, stalls and epochs

With input acceptance at E0, beam output is visible after E7 and first transferable at E8. Global stalls hold all eight stages. AEC output is visible after E(63*T+3), including 56 divider iterations per tap; T=32 gives2019 clocks. The next AEC transaction waits for result transfer. The fork adds registered storage. VAD is visible after E2 following last-bin acceptance; logarithm has eight stallable stages. Commit waits for accepted work to retire, swaps delays, captures enables and clears histories, weights, powers, partial frames and hangover. Incomplete frames are discardable; withdrawn requests do not commit. Reset aborts pending work and selects zero delays, AEC enabled and gating enabled.

F Legacy preservation and storage ownership

All eighteen inherited RTL modules remain byte-identical to the reference inputs. Chain modes retain bypass, FIR, SOS, FIR→SOS and SOS→FIR; feature modes retain audio, SRC audio, spectral tokens and external-power tokens. Beamformer history uses one synchronous writer, avoiding ambiguous ownership from generated dynamically indexed writers while preserving pre-write operand capture, validity-based initialization and pipeline behavior. Source: E023. Current successful physical mapping and timing are established separately by E078.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent arbitrary-precision Python oracle against Icarus and compiled Verilator RTL; mathematically reconstructed constants checked separately against ROMs.

Result. Exact tested audio, freeze, spectrum, Mel-energy and token comparisons passed. Constants covered512 Hann values,1024 twiddle components,16448 Mel weights and17 log endpoints. No physical FPGA acoustic numerical comparison was performed.

Evidence: E044, E034, E068, E069, E070, E071

B Inherited compatibility

Method. Five covering configurations, five chain modes, four channels, all64 SRC L/M pairs, full inherited token processing and malformed-frame recovery.

Result. Passed; six spectral frames compared1542 bins, Mel tests512 bands, and logarithm tests517 cases in each simulator.

Evidence: E068, E033

C NLMS convergence and recovery

Method. Seeds17,29,43 each exercised T=1,8,16,32 with16384 training,4096 evaluation,1024 double-talk, 2048 recovery and4096 re-evaluation samples.

Result. Evaluation/recovery ERLE, in tap order1,8,16,32: seed17 64.51/64.13,66.84/66.78,63.52/63.41,58.43/58.46 dB; seed29 64.53/64.11,66.92/66.78,63.49/63.38,58.52/58.38 dB; seed43 64.22/64.31,66.95/67.01,63.63/63.76,58.70/58.68 dB. All12 cases exceeded20/18 dB thresholds.

Evidence: E069, E070, E071

D Double-talk and leakage sensitivity

Method. Public audio/freeze scoreboards during near-end injection, reference silence and model-defined nonzero-weight leakage phases; separate removed-leakage mutant.

Result. All12 final512-sample double-talk windows reported100% freezing, exceeding95%, with exact audio/freeze agreement. Leakage-phase output comparisons passed and removed leakage caused numerical rejection. Model weight statistics were not direct measurements of DUT weight identity or individual tap updates.

Evidence: E069, E070, E071, E073, E043

E Spatial improvement

Method. 4096 post-warmup samples; separate real DUT simulations for desired/noise and desired/interferer contributions. Steering used0.04 m spacing,343 m/s and±30°.

Result. Broadside SNR gains were9.228,9.069,8.921 dB versus>=7 dB. Steering SIR improved22.679/+30° and22.678/−30° dB versus>=6 dB; desired gains−0.1043/−0.1046 dB were within1 dB of unity.

Evidence: E069, E070, E071

F VAD, tokens and throughput

Method. 18 complete frames per seed across three epochs with independent stalls and concurrent inherited traffic; four-frame continuously ready T=32 measurement.

Result. Each seed produced18 events and704 tokens: seven suppressed frames among the first twelve, then six ungated frames. The measured window accepted2048 tuples over4876126 clocks,42000.56 tuples/s at100 MHz versus>=16000. Maximum acceptance gap was131110 clocks.

Evidence: E069, E070, E071, E028

G Boundary and protocol robustness

Method. Legal covering matrix, twelve illegal settings, signed ties/saturation, VAD thresholds, division resets, epoch withdrawal and independent stalls.

Result. Passed; all four tap counts executed division-reset cases. Depths1,17,32 exercised partial-frame commits/resets; illegal examples produced required diagnostics.

Evidence: E072, E048

H Defect sensitivity

Method. Seven disposable RTL mutants compiled and executed against unchanged numerical scoreboards.

Result. 7/7 rejected: fraction complement, microphone omission, gradient sign, removed double-talk protection, removed leakage, VAD threshold and token multiplier corruption. Four unmodified controls passed.

Evidence: E073, E042

I Generic synthesis

Method. Yosys hierarchy/consistency checks and memory-preserving synthesis of default, singleton and maximum acoustic configurations.

Result. Substantive inherited/acoustic structure retained; zero latches, unresolved black boxes or division/modulo cells. Generic cell totals7914/7729/8109 are not FPGA area.

Evidence: E074, E053

J Portable reproduction

Method. Fresh copied declared sources with fresh Icarus, Verilator and Yosys outputs.

Result. Passed complete512-sample acoustic-frame comparisons with one event and64 tokens in each simulator, compact legacy checks and zero-latch synthesis. This was a bounded subset, not another full campaign.

Evidence: E075, E046

K L1 RTL timing lint

Method. Native analysis of24 delivered RTL files.

Result. Passed with zero combinational divider/modulo, mixed-assignment, LUT clock-gating or loop-unroll hazards.

Evidence: E077

L L2 physical timing and resources

Method. Intel Quartus Prime17.1 TimeQuest post-fit STA for default billet_f5_top on Arria10 GX at10.0 ns using virtual pins.

Result. Passed: WNS_setup = +1.808 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 122.07 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E078

M L3 physical Arria10 FPGA HIL

Method. Three consecutive physical PCIe BAR0 bridge register probes.

Result. 3/3 deterministic passes. Connectivity-only scope; no F5 acoustic workload vectors or on-board numerical comparisons.

Evidence: E079

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	13 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 29	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

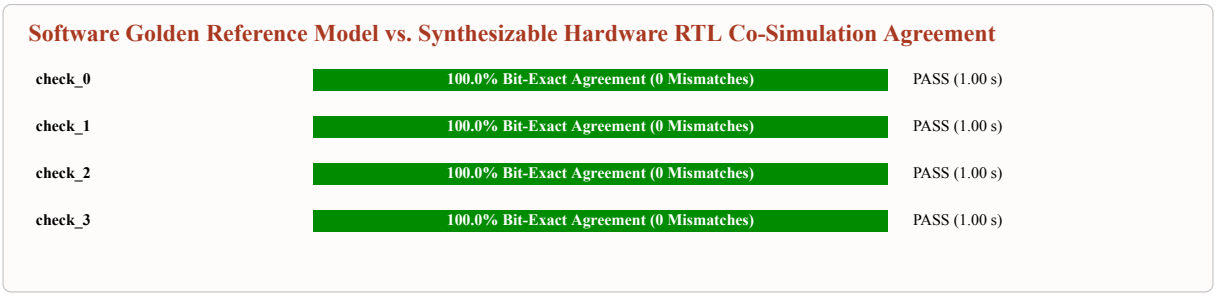


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.808 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

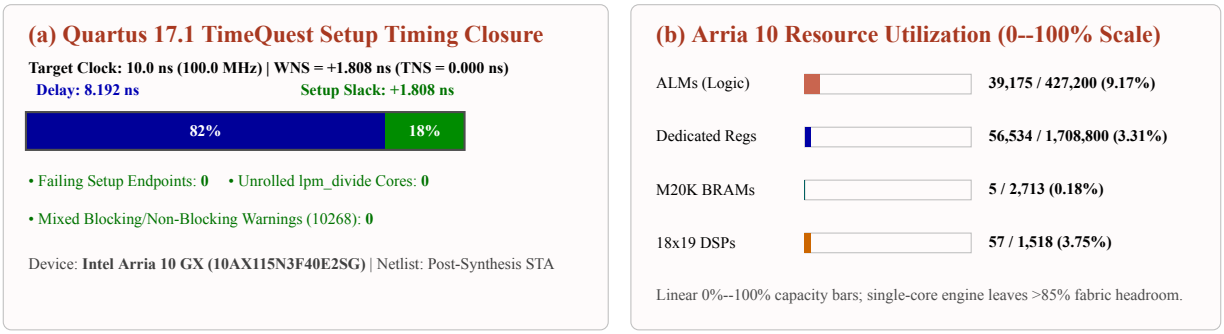


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

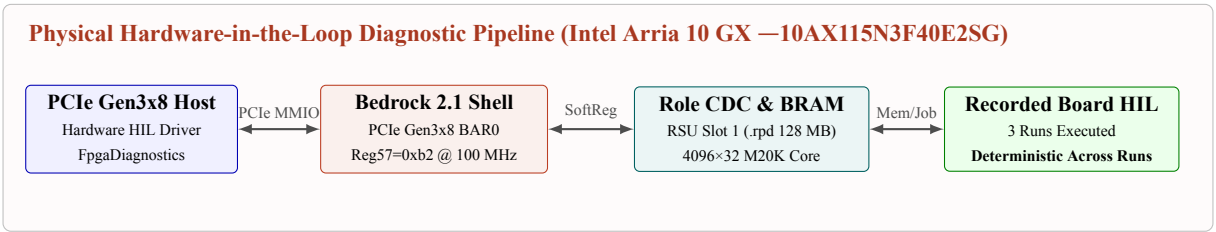


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top billet_f5_top. design/SPEC.md defines numerical and interface rules; design/ARCHITECTURE.md describes registered boundaries; design/GENERATIONAL_HANDOFF.md indexes read-only parent inputs.

Assert reset, release it, and hold valid payloads stable while stalled. Pack eight synchronized signed samples and their matching far-end reference into each tuple. Buffer externally acquired PCM for bursty acceptance.

Handshake all eight inactive delay entries, then hold a_commit_valid and desired enables until acceptance while consumers drain. Consume VAD events even when tokens are suppressed.

Dependencies are Python3 standard library, Icarus/vvp, Verilator with a C++ toolchain, and Yosys. From the project root run python3 tests/verify.py --suite compatibility; acoustic runs use python3 tests/verify.py --suite acoustics --seed 17, with separate seeds29 and43. Run the same entrypoint for suites robustness, defect-sensitivity, synthesis and portable.

Each invocation creates a fresh evidence/l1-<suite>-<identifier>/ directory with receipt.json, commands, hashes and retained results. Native physical timing and bridge gates are separate from portable L1 execution.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 24 Verilog/SystemVerilog files; 3850 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_f2_coeff_bank.sv	70 total	70 code
rtl/billet_f2_history.sv	43 total	43 code
rtl/billet_f2_mac.sv	110 total	104 code
rtl/billet_f2_round_sat.sv	32 total	28 code
rtl/billet_f2_top.sv	149 total	141 code
rtl/billet_f3_round_sat.sv	29 total	26 code
rtl/billet_f3_sos_coeff_bank.sv	79 total	75 code
rtl/billet_f3_sos_engine.sv	264 total	257 code
rtl/billet_f3_top.sv	252 total	243 code
rtl/billet_f4_hann_rom.sv	522 total	520 code
rtl/billet_f4_log.sv	128 total	128 code
rtl/billet_f4_log_rom.sv	26 total	24 code
rtl/billet_f4_mel.sv	132 total	130 code
rtl/billet_f4_mel_rom.sv	272 total	270 code
rtl/billet_f4_spectrum.sv	178 total	174 code
rtl/billet_f4_src.sv	206 total	203 code
rtl/billet_f4_top.sv	201 total	199 code
rtl/billet_f4_twiddle_rom.sv	525 total	523 code
rtl/billet_f5_audio_fork.sv	41 total	41 code
rtl/billet_f5_beamformer.sv	141 total	141 code
rtl/billet_f5_nlms.sv	167 total	167 code
rtl/billet_f5_top.sv	245 total	242 code
rtl/billet_f5_udiv.sv	39 total	39 code
rtl/billet_f5_vad.sv	62 total	62 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E023	rtl/billet_f5_beamformer.sv
E025	rtl/billet_f5_top.sv
E028	tests/acoustic_top_tb.sv
E033	tests/compatibility.py
E034	tests/constants.py
E042	tests/mutations.py
E043	tests/nlms_tb.sv
E044	tests/oracle.py
E046	tests/portable.py
E048	tests/robustness.py
E053	tests/synthesis.py
E068	.vulcan/engineering/run-0001/attempt-0022/stdout.log
E069	.vulcan/engineering/run-0001/attempt-0023/stdout.log
E070	.vulcan/engineering/run-0001/attempt-0024/stdout.log

Reference	Original project-relative record
E071	.vulcan/engineering/run-0001/attempt-0025/stdout.log
E072	.vulcan/engineering/run-0001/attempt-0026/stdout.log
E073	.vulcan/engineering/run-0001/attempt-0027/stdout.log
E074	.vulcan/engineering/run-0001/attempt-0028/stdout.log
E075	.vulcan/engineering/run-0001/attempt-0029/stdout.log
E077	.vulcan/engineering/run-0001/rtl-timing-lint.json
E078	.vulcan/engineering/run-0001/quartus-sta.json
E079	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).